# DOSSIER

## Product Design & Mission Architecture Document

AI-Powered Intelligence Training Platform

| | |
|---|---|
| **Version** | 1.0 — Initial Release |
| **Status** | Active Development |
| **Classification** | Internal — AI Builder Reference |
| **Last Updated** | 2025 |
| **Stack** | Next.js · Tailwind · Framer Motion · Gemini · ElevenLabs |

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# 1. Product Overview

Dossier is an AI-powered interactive intelligence training platform. Users complete structured missions designed to build three critical skills for the modern information age: detecting AI-generated misinformation, conducting psychology-based interrogations, and defending classified systems from cyberattacks.

The experience is cinematic, adaptive, and high-stakes. Every mission is procedurally generated by AI — no two runs are identical. Dossier is built to feel like a billion-dollar intelligence agency, not a gamified quiz app.

### CORE PILLARS

- Adaptive AI — missions evolve in real time based on the operative's behavior and performance
- Immersion — voice-acted suspects, live dashboards, cinematic UI, procedural scenarios
- Legitimacy — rooted in real intelligence tradecraft, behavioral psychology, and cybersecurity
- Replayability — AI ensures no two missions are identical; difficulty scales with skill

### TARGET AUDIENCE

| Segment | Description |
|---|---|
| **Primary** | Tech-forward adults 18–35 who consume spy thrillers, cybersecurity content, or true crime |
| **Secondary** | Corporate security teams, journalism students, researchers in media literacy |
| **Aspirational** | Intelligence agencies, universities, military training programs |

# 2. Technical Stack

| Layer | Technology |
|---|---|
| **Frontend** | Next.js 14 (App Router) + Tailwind CSS + Framer Motion + shadcn/ui |
| **AI Engine** | Google Gemini 1.5 Pro — mission generation, NPC behavior, adaptive scoring |
| **Voice** | ElevenLabs — voice synthesis for suspect NPCs and mission briefings |

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| Layer | Technology |
|---|---|
| **Auth** | Clerk or Supabase Auth — operative profiles, clearance levels, progress tracking |
| **Database** | Supabase (PostgreSQL) — mission logs, scores, operative data |
| **State** | Zustand — global mission state, session management |
| **Deployment** | Vercel — edge functions for AI calls, streaming responses |

### AI INTEGRATION NOTES

- All mission content is generated via Gemini with structured JSON output schemas (see Section 5)
- Voice lines are generated per-session via ElevenLabs API — cache aggressively to reduce latency
- Use streaming responses for interrogation dialogue to simulate real-time conversation
- Implement a fallback static scenario bank for API outages

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# 3. Mission Architecture

Each mission in Dossier follows a shared structural framework called the Mission Loop. All three mission types implement this loop, with type-specific phases layered on top.

## 3.1 The Universal Mission Loop

| Phase | Description |
|---|---|
| **Phase 1: Briefing** | Operative receives a classified dossier. AI generates scenario background, objectives, and constraints. Voice-acted by ElevenLabs handler character. |
| **Phase 2: Investigation** | Core mission gameplay. Type-specific — see Sections 4.1–4.3. |
| **Phase 3: Decision** | Operative makes a final judgment call based on gathered intelligence. Binary or multi-choice. |
| **Phase 4: Debrief** | AI scores the mission. Reveals hidden information. Explains what was missed and why. |
| **Phase 5: Progression** | XP awarded. Clearance level updated. Next mission unlocked or difficulty escalated. |

## 3.2 Difficulty & Clearance System

Dossier uses a 5-tier clearance system that gates mission complexity and AI adversary sophistication:

| Clearance | Characteristics |
|---|---|
| **Level 1 — RECRUIT** | Obvious tells, single threat vector, guided hints available |
| **Level 2 — FIELD AGENT** | Subtle inconsistencies, 2 threat vectors, no hints |
| **Level 3 — OPERATIVE** | Layered deception, contradictory evidence, time pressure |
| **Level 4 — SPECIALIST** | AI actively adapts to operative behavior mid-mission |
| **Level 5 — HANDLER** | Full adversarial AI, multiple simultaneous threats, no debrief until series complete |

**BUILDER NOTE**

For AI builders: always pass the current clearance_level and operative_history into the mission generation prompt. The AI must use these to calibrate adversary sophistication, evidence complexity, and number of red herrings.

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# 4. Mission Types

## 4.1 Mission 01 — Deepfake Detection

### OBJECTIVE

The operative receives a package of media artifacts — videos, images, audio recordings, and documents. Some are real. Some are AI-generated fakes seeded with embedded misinformation. The operative must identify which are synthetic and expose the disinformation campaign before it reaches the public.

### GAMEPLAY LOOP

1. AI generates a scenario: e.g. a fabricated political leak, a fake scientific report, a synthetic eyewitness video
2. Operative is presented with 5–12 media artifacts and contextual documents
3. Operative tags each artifact as REAL, SYNTHETIC, or UNCERTAIN
4. Operative submits a final threat assessment naming the disinformation campaign's objective
5. AI scores based on accuracy, confidence calibration, and narrative coherence of the assessment

### AI PROMPT SCHEMA — MISSION GENERATION

```
POST /api/mission/generate
{
  "mission_type": "deepfake_detection",
  "clearance_level": 1-5,
  "scenario_seed": "string (optional — forces theme)",
  "operative_history": [...previous_mission_results],
  "output_schema": {
    "scenario_title": "string",
    "briefing_text": "string (2-3 paragraphs)",
    "artifacts": [
      {
        "id": "string",
        "type": "video|image|audio|document",
        "title": "string",
        "content_description": "string",
        "is_synthetic": boolean,
        "tell_type": "compression|metadata|inconsistency|none",
        "difficulty": 1-5
      }
    ],
    "campaign_objective": "string (hidden — revealed in debrief)",
    "red_herrings": ["string"],
    "debrief_explanation": "string"
  }
```

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```
}
```

#### SCORING DIMENSIONS

| Dimension | Weight |
|---|---|
| **Accuracy** | 50% |
| **Confidence Calibration** | 20% |
| **Assessment Quality** | 20% |
| **Speed Bonus** | 10% |

#### UI COMPONENTS REQUIRED

- MediaViewer — renders artifact with zoom, metadata panel, and anomaly highlighting
- TagSelector — REAL / SYNTHETIC / UNCERTAIN toggle per artifact
- ThreatAssessmentInput — free-text field with AI-assisted completion
- EvidenceBoard — drag-and-drop connection map linking related artifacts
- DebriefReveal — animated overlay revealing truth per artifact with color-coded scoring

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## 4.2 Mission 02 — Interrogation Training

#### OBJECTIVE

The operative conducts a live interrogation of an AI-powered NPC suspect. The suspect has a hidden backstory, a secret they are concealing, and a set of behavioral patterns rooted in real psychology. The operative must extract the truth using strategic questioning — detecting contradiction, exploiting cognitive bias, and reading behavioral signals — without tipping the suspect off.

#### GAMEPLAY LOOP

6. AI generates a suspect profile: backstory, secret, psychological profile, baseline behavior
7. Operative receives a pre-interrogation dossier with partial information
8. Live conversation begins — operative types questions, suspect responds via Gemini + ElevenLabs
9. Operative tracks detected contradictions and behavioral flags on a sidebar panel
10. Operative submits a final verdict: what the suspect is hiding and the key evidence
11. AI scores based on information extracted, contradictions caught, and tactics used

#### AI SUSPECT BEHAVIOR RULES

- Suspect must maintain internal consistency — all answers must reference a hidden source-of-truth JSON

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- Suspect defensiveness escalates if operative is too direct too early
- Suspect will plant misdirections if they detect the operative is close to the truth
- Suspect has 3 vulnerability windows per session — moments where they are more likely to slip
- Suspect uses real psychological defense mechanisms: deflection, projection, minimization

### AI PROMPT SCHEMA — SUSPECT GENERATION

```
POST /api/suspect/generate
{
  "clearance_level": 1-5,
  "scenario_seed": "string (optional)",
  "output_schema": {
    "name": "string",
    "age": number,
    "occupation": "string",
    "surface_story": "string",
    "hidden_truth": "string (NEVER revealed to operative)",
    "psychological_profile": {
      "type": "narcissist|deflector|minimizer|projector",
      "tells": ["behavioral signal strings"],
      "vulnerability_triggers": ["string"]
    },
    "known_facts": ["string (given to operative)"],
    "contradictions_to_plant": ["string"],
    "voice_id": "elevenlabs_voice_id"
  }
}
```

### AI PROMPT SCHEMA — LIVE INTERROGATION TURN

```
POST /api/suspect/respond
{
  "suspect_profile": {...full_suspect_object},
  "conversation_history": [...turns],
  "operative_question": "string",
  "tension_level": 0-100,
  "vulnerability_window_active": boolean,
  "output_schema": {
    "response_text": "string",
    "behavioral_flag": "string|null",
    "contradiction_introduced": boolean,
    "tension_delta": number
  }
}
```

### UI COMPONENTS REQUIRED

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- SuspectPortrait — animated character face with micro-expression system (CSS keyframes)
- InterrogationChat — streaming dialogue with typing indicators
- TensionMeter — real-time 0-100 bar showing suspect stress level
- ContradictionLog — sidebar where operative manually flags and annotates inconsistencies
- PsychProfile — partially revealed sidebar card updating as operative gathers intel
- VerdictForm — structured form: What are they hiding? Key evidence? Recommended action?

## 4.3 Mission 03 — Cyber Defense

### OBJECTIVE

The operative is the last line of defense for a classified server infrastructure. An AI adversary is actively probing, breaching, and escalating access across the network. The operative must identify intrusion vectors, contain breaches in real time, patch vulnerabilities, and neutralize the attack before critical data is exfiltrated.

### GAMEPLAY LOOP

12. AI generates a network topology: servers, endpoints, data nodes, and vulnerability map
13. Attack sequence begins — intrusions surface progressively, not all at once
14. Operative responds by: isolating compromised nodes, deploying countermeasures, patching CVEs
15. AI adversary adapts tactics based on operative's responses — pivots if blocked
16. Mission ends on: full containment (success), data exfiltration (partial fail), or system collapse (fail)

### AI ADVERSARY BEHAVIOR RULES

- Adversary follows a kill chain: Reconnaissance → Initial Access → Persistence → Lateral Movement → Exfiltration
- Each clearance level adds one additional concurrent attack thread
- At Level 4+, adversary plants false alerts to distract operative from real threats
- Adversary learns from operative's containment patterns and adjusts within the same session

### AI PROMPT SCHEMA — NETWORK GENERATION

```
POST /api/mission/cyber/generate
{
  "clearance_level": 1-5,
  "output_schema": {
    "network_name": "string",
```

```
    "nodes": [
      {
        "id": "string",
        "type": "server|endpoint|database|gateway",
        "name": "string",
        "vulnerabilities": ["CVE-XXXX-XXXX"],
        "data_sensitivity": 1-5,
        "connections": ["node_id"]
      }
    ],
    "attack_sequence": [
      {
        "tick": number,
        "action": "probe|breach|persist|pivot|exfil",
        "target_node": "node_id",
        "visible_to_operative": boolean,
        "countermeasure": "patch|isolate|block|honeypot"
      }
    ],
    "decoy_alerts": [...],
    "critical_data_nodes": ["node_id"]
  }
}
```

### UI COMPONENTS REQUIRED

- NetworkGraph — D3.js or Framer Motion node graph, real-time status coloring per node
- ThreatFeed — scrolling log of detected events with timestamps
- ActionPanel — contextual action menu per selected node: Isolate / Patch / Monitor / Honeypot
- AttackTimeline — horizontal kill chain progress bar showing adversary stage
- DataExfilMeter — fills as adversary approaches critical data — creates urgency
- PostMortemMap — debrief overlay showing full attack path and all missed events

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# 5. Operative Profile System

Every user has a persistent Operative Profile that tracks progression, specializations, and behavioral patterns across all mission types.

### PROFILE SCHEMA

```
{
  "operative_id": "uuid",
  "callsign": "string",
  "clearance_level": 1-5,
  "xp": number,
  "missions_completed": number,
  "specializations": {
    "deepfake_detection": 0-100,
    "interrogation": 0-100,
    "cyber_defense": 0-100
  },
  "behavioral_patterns": {
    "avg_question_directness": 0-1,
    "risk_tolerance": 0-1,
    "speed_vs_accuracy_ratio": 0-1
  },
  "mission_history": [
    {
      "mission_id": "uuid",
      "type": "deepfake|interrogation|cyber",
      "score": 0-100,
      "clearance_level": 1-5,
      "duration_seconds": number,
      "key_decisions": [...],
      "timestamp": "ISO8601"
    }
  ]
}
```

**BUILDER NOTE**

Pass the full operative_history array into every mission generation call. The AI uses this to: avoid repeating scenario themes, escalate difficulty appropriately, and tailor adversary tactics to the operative's known weaknesses.

# 6. Scoring & Progression

| Score Range | Rating |
|---|---|
| **Score Range** | Rating |
| **90–100** | ELITE — Field Ready |
| **75–89** | PROFICIENT — Cleared for Deployment |
| **55–74** | DEVELOPING — Additional Training Required |
| **0–54** | COMPROMISED — Mission Failed |

XP awards scale with both performance and clearance level difficulty multiplier:

```
XP = base_score × clearance_multiplier × speed_bonus
clearance_multiplier: [1.0, 1.3, 1.7, 2.2, 3.0]
speed_bonus: max(1.0, 1.5 - (duration / par_time))
```

# 7. Design System

### COLOR PALETTE

| Token | Value |
|---|---|
| **Background** | #050505 — near black |
| **Panel** | #111111 — dark surface |
| **Accent** | #00FF9D — emerald green |
| **Text Primary** | #F0F0F0 |
| **Text Secondary** | #888888 |
| **Text Muted** | #444444 |
| **Border** | #1E1E1E |

### TYPOGRAPHY

| Role | Font |
|---|---|
| **Display / Headlines** | Syne — weight 800, tracking -0.02em |
| **Body** | Inter — weight 300–400, line-height 1.7 |
| **Labels / Mono** | JetBrains Mono — weight 400–500, tracking 0.1–0.15em |

### ANIMATION PRINCIPLES

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- All reveals: fadeSlideUp — opacity 0→1, translateY 24px→0, duration 0.7s, ease
- Hover states: subtle glow border rgba(0,255,157,0.3), background rgba(0,255,157,0.04)
- Loading states: scan line animation — horizontal emerald sweep, 3s loop
- Tension elements: pulse animation on status dots, radar sweep at 3s loop
- Page transitions: Framer Motion layout animations, shared element transitions between mission phases

# 8. AI Builder Checklist

Use this checklist when building any mission. Every item must be implemented before a mission is considered complete.

### Mission Generation

- Gemini prompt includes clearance_level and full operative_history
- Output validated against mission JSON schema before rendering
- Fallback static scenario available if API fails
- Scenario theme deduplication against last 5 missions

### Voice & Audio

- Briefing voiceover generated via ElevenLabs before mission starts (not on-demand)
- Interrogation responses streamed and synthesized per turn — max 2s latency target
- Voice cached per session, not regenerated on replay
- Audio fails gracefully — text always readable without voice

### Scoring & Debrief

- Score calculated server-side — never trust client score submission
- Debrief reveals all hidden information the operative missed
- Mission result logged to Supabase with full decision trace
- Operative profile updated atomically after score commit

### Adaptive AI

- Adversary / suspect behavior references operative's historical patterns
- Mid-mission adaptation does not retroactively change established facts
- AI cannot make mission impossible — minimum 1 correct path always exists
- All AI outputs filtered for content safety before display

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**DESIGN PRINCIPLE**

When in doubt: the mission must feel real. Every design decision — UI layout, AI response time, voice acting, scoring logic — should serve immersion first. If something feels like a quiz, it needs to be rebuilt.

— END OF DOCUMENT —

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